Assignment 3: Revenue of Intercollegiate Athletics Programs in the US

FEM11149 - Introduction to Data Science

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1. Introduction

Intercollegiate athletic programs are an important part of higher education in the United States, and running such a program requires substantial financial resources. However, not every program generates the same amount of revenue. While some larger athletic programs may earn over $100 million USD per year, many programs at smaller colleges have less than $1 million USD available for their teams. For college athletic directors, it is therefore useful to know what revenue can be expected for a program with certain characteristics. This may aid them with planning their budget and comparing their program to similar institutions. Colleges and universities that participate in federal student aid programs have to report information about their athletic programs to the US Department of Education (USDE), for example about the number of college athletes, as well as the coaches and their respective salaries. Many of these characteristics are closely related, since larger programs usually have more athletes and more coaches at the same time. It is therefore relevant to investigate whether this information can be used to estimate the revenue of a program, and whether a few general features of a program already contain as much information as all characteristics together. Using the reported information from the USDE, the aim of this report is to answer two research questions. To begin with, **to what extent can the revenue of an intercollegiate athletic program be predicted from a few underlying program characteristics, compared with using all characteristics separately?** Secondly, based on this, **which characteristics, either on their own or combined into these underlying features, are most closely related to revenue?** Since this is a first exploratory analysis, the focus is on prediction and on relationships with revenue, without making claims about what causes differences in revenue.

2. Data

The data come from the Equity in Athletics Disclosure Act (EADA) survey of the US Department of Education. Every co-educational college or university that participates in federal student aid programs and has at least one intercollegiate athletic program annually reports information about its programs (USDE, 2026). The dataset used for this report contains 1,000 institutions. Two variables are categorical: the athletic association and division of the institution (19 levels, e.g. NCAA Division I-FBS) and its sector (5 levels, e.g. private nonprofit, 4-year or above). The other 28 variables, which are used as predictors, describe undergraduate enrollment, the number of athletes and teams, the participants per sport, the number and average salaries of head and assistant coaches, and the shares of full-time and female head coaches, mostly separately for men and women. The number of athletes counts each athlete once, while the participants per sport count an athlete once for every team they are on (USDE, 2026). The unit ID identifies the institution and is not used as a predictor. The variable of interest is the total revenue of the athletic program, which includes, for example, ticket sales, media rights, donations and support from the institution itself. Since revenue has to cover the expenses of the program, a deficit paid by the institution also counts as revenue (USDE, 2025). Revenue is very right-skewed: it ranges from $13,766 to $289.6 million, with a median of

$4.1 million and a mean of $14.3 million (Appendix, Figure 1). Therefore, the logarithm of revenue, which is much more symmetric, is used as the dependent variable.

To evaluate the predictions on data that are not used for estimation, the 1,000 institutions are randomly split into a training set of 800 institutions (80%) and a test set of 200 institutions (20%). The split is stratified on log revenue, so that both sets contain small as well as large programs.

Missing values only occur in the coaching variables. Institutions without men's or women's teams report no coaches for those teams, so the number of coaches and their salaries are set to zero. For 10 institutions, assistant coaches are reported without a salary (16 missing salary values in total). As volunteer coaches are left out of the average salary (USDE, 2026), these may be programs with only volunteers. However, since this cannot be verified, the missing salaries are replaced by the median of the training data. The coach counts, the other counts and the salaries are also right-skewed, so they are transformed with $\log(1+x)$, which keeps zeros defined and reduces the influence of very large programs.

A second dataset contains 10 institutions for which the revenue is predicted. Their unit IDs are replaced by the numbers 1 to 10.

3. Methods

The first technique is Principal Component Analysis (PCA). PCA tries to describe the variables in a dataset as well as possible by a few new, uncorrelated variables, called principal components. This is useful when variables are strongly correlated and therefore largely measure the same thing. Each component is a weighted sum of the original variables and can often be interpreted as a general feature of the observations, such as their size. PCA is unsupervised: it only uses the predictors, not the dependent variable.

Let $\mathbf{X}$ be the $n \times p$ matrix of predictors $X_1, \dots, X_p$, with n observations and p predictors. Component s is the linear combination $Z_s = \gamma_{1s}X_1 + \cdots + \gamma_{ps}X_p = \gamma_s^\top \mathbf{X}$, where γ_s is the loading vector with the weights of component s. The first component has the largest possible variance:

$$\max_{\gamma_1 : \|\gamma_1\|=1} \mathrm{Var}(\gamma_1^\top \mathbf{X}) = \max_{\gamma_1 : \|\gamma_1\|=1} \gamma_1^\top \Sigma \gamma_1,$$

where $\Sigma = \mathrm{Var}(\mathbf{X})$ is the covariance matrix, and the restriction $\|\gamma_1\| = 1$ prevents an infinitely large variance. The solution is the first eigenvector of Σ, and $\mathrm{Var}(Z_1)$ is the first eigenvalue λ_1. Each next component again has maximum variance, but is uncorrelated with the previous ones, which gives $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$. Since PCA is sensitive to the scale of the variables, it is done on the correlation matrix, which is the same as using z-scores. The eigenvalues then add up to p, and the proportion of variance accounted for by component s is $\mathrm{VAF}_s = \lambda_s / p$.

To decide how many components r to keep, four methods are used. First, the cumulative VAF should be at least 80%, although this can lead to many components. Second, Kaiser's rule keeps components with $\lambda_s > 1$, since a component equal to one standardised variable would have $\lambda_s = 1$. Third, in the scree plot of λ_s against s, the components are retained up to the elbow, without the elbow itself. Fourth, the permutation test compares each λ_s to the eigenvalues of $N_{\mathrm{perm}} = 1000$ datasets in which the values within each column are randomly permuted, which removes the correlations between the predictors. If λ_s lies above the $1 - \alpha/2$ percentile of the permuted eigenvalues, with $\alpha = 0.05$, dimension s is significantly different from random data and is kept.

The stability of the eigenvalues and the VAF is assessed with the bootstrap. $B = 1000$ datasets $\mathbf{X}^{*b}$ of size $n \times p$ are drawn by sampling rows of $\mathbf{X}$ with replacement, and in each the eigenvalues λ_s^{*b} are computed. Their 2.5 and 97.5 percentiles give a 95% confidence interval, where a narrow interval means a stable result. An eigenvalue is significantly larger than 1 (Kaiser's rule) if the whole interval lies above 1, and an interval is also constructed for the cumulative VAF of the selected components.

To interpret the components, the component loadings are used: the correlations between the variables and the component scores, $\rho_{js} = \mathrm{cor}(X_j, Z_s) = \gamma_{js}\lambda_s^{1/2}$ for standardised variables. They lie between -1 and $+1$, and large positive or negative values show which variables a component represents, so that it can be named.

A biplot shows the observations as points and the variables as arrows for the first two components. The fit per variable, $\sum_{s=1}^{r} \rho_{js}^2$, is the proportion of the variance of X_j accounted for by the first r components; a value close to 1 means the variable is well represented.

In the principal component regression (PCR), the dependent variable is regressed with Ordinary Least Squares (OLS) on the first r component scores and dummy variables for the categorical predictors:

$$y_i = \beta_0 + \sum_{s=1}^{r} \theta_s z_{is} + \sum_{l=1}^{L} \delta_l d_{il} + \varepsilon_i,$$

where y_i is the dependent variable for observation i, β_0 the intercept, z_{is} the score on component s with coefficient θ_s, d_{il} the l-th of L dummies with coefficient δ_l, and ε_i the error term. For each component, the following hypotheses are tested:

$$H_0 : \theta_s = 0 \quad \text{versus} \quad H_1 : \theta_s \neq 0.$$

The null hypothesis H_0 states that component s has no relation with the dependent variable, given the other components and the dummies, while the alternative hypothesis H_1 states that it does. The test uses the t-value $t = \hat{\theta}_s / \mathrm{SE}(\hat{\theta}_s)$, which is the estimate divided by its standard error. The p-value is the probability of finding a t-value at least this large in absolute value if H_0 is true. If $p < 0.05$, which in large samples corresponds to $|t| > 1.96$, H_0 is rejected and the component is significant at the 5% level. A higher R^2 is preferred, as more variance of the dependent variable is then explained by the model.

The second technique is ridge regression, an adaptation of multiple regression that penalises the size of the coefficients. It is very useful with many or strongly correlated predictors, because the estimates become more stable. With the dependent variable in deviation from its mean, ridge regression minimises

$$L(\beta_1, \ldots, \beta_q) = \underbrace{\sum_{i=1}^{n} \Big(y_i - \sum_{j=1}^{q} x_{ij} \beta_j \Big)^2}_{\text{regression term}} + \underbrace{\lambda \sum_{j=1}^{q} \beta_j^2}_{\text{penalty term}},$$

where q is the number of predictors including dummies, x_{ij} the value of predictor j for observation i, β_j its regression weight, and $\lambda > 0$ the penalty parameter (not the eigenvalues λ_s of the PCA). Ridge regression shrinks the β_j towards zero, and λ controls the strength of this shrinkage: as $\lambda \to 0$ the estimates become the OLS estimates, and as $\lambda \to \infty$ they go to zero. The ridge estimates always exist, even under multicollinearity. Because all weights are penalised equally, the numerical predictors are standardised to z-scores. Ridge regression is preferred over the LASSO, as the LASSO can set coefficients to zero and tends to keep only one of several strongly correlated predictors, whereas ridge regression keeps all of them.

The value of λ is chosen from a grid with K-fold cross-validation ($K = 10$): each fold k is left out once, the model is estimated on the other folds, and the n_k observations in fold k are predicted, which gives

$$\mathrm{MSE}_k = \frac{1}{n_k} \sum_{i \in \text{fold } k} (y_i - \hat{y}_i)^2, \qquad \mathrm{RMSE}_{\mathrm{CV}} = \sqrt{\frac{1}{K} \sum_{k=1}^{K} \mathrm{MSE}_k},$$

where $\hat{y}_i$ is the out-of-sample prediction for observation i. A lower $\mathrm{RMSE}_{\mathrm{CV}}$ is better. Both $\lambda_{\min}$, with the lowest $\mathrm{RMSE}_{\mathrm{CV}}$, and λ_{1SE}, the largest λ within one standard error of this minimum, are considered; the latter gives a simpler model with more shrinkage.

Both models are estimated on the training set and compared on the test set with $\mathrm{RMSE} = \sqrt{\frac{1}{m} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2}$ and $R^2 = 1 - \sum_i (y_i - \hat{y}_i)^2 / \sum_i (y_i - \bar{y})^2$, where m is the number of test observations and $\bar{y}$ their mean. A lower RMSE and a higher R^2 mean better predictions. For a log-transformed dependent variable, errors on the log scale can be read as relative errors. Predictions are transformed back to the original scale with $\exp(\hat{y}_i)$, which estimates the median rather than the mean, since $E[\exp(y)] > \exp(E[y])$. On the original scale, the RMSE is mostly determined by the largest observations.

4. Results

The first principal component has an eigenvalue of 13.80 and explains 49.3% of the variance of the 28 predictors (Appendix, Table 1). The selection methods do not all agree with each other. The 80% VAF rule suggests 8 components, Kaiser's rule 6, the bootstrapped Kaiser's rule 5, the scree criterion two (the eigenvalues drop sharply after the second component), and the permutation test 3. Since the permutation test is the only method that compares the eigenvalues to those of random data, 3 components are selected. Components 4 to 6 have eigenvalues above 1, but not above what can be expected by chance. Together, the 3 components explain 64.4% of the variance, and the narrow bootstrap interval of [63.1%, 66.5%] shows that this result is stable.

The loadings (Appendix, Table 2) and the biplot (Appendix, Figure 2) show what the components mean. The first component correlates positively with all variables, most strongly with the number of athletes, teams and coaches, so it measures *program size*. The second component is high for programs at large institutions with high coach salaries and many full-time coaches, and low for programs with many teams and coaches but lower salaries. It can be called *pay versus breadth*. The third component separates track and field and other individual sports from basketball, baseball and softball. Most variables are well represented by the three components, except the share of female head coaches and the number of football players.

In the PCR, the components and the dummies explain 94.5% of the variance of log revenue. Program size and pay versus breadth are significant at the 5% level ($t = 44.1$ and $t = 9.5$, both $p < 0.001$), but the third component is not ($t = 0.47$, $p = 0.64$). The division is also very important: an NCAA Division I-FBS program earns about 11 times as much as a CCCAA program with the same component scores. For ridge regression, cross-validation chooses a very small penalty ($\lambda_{\text{min}} = 5.0\text{e-}04$), so the results are close to OLS. The largest ridge coefficients are again the division dummies, followed by the salary of head coaches of men's teams.

On the test set (Appendix, Table 3), both models predict log revenue well. The PCR has the lowest RMSE on the log scale (0.425, $R^2 = 0.916$), so three components and the division contain almost the same information as all 28 characteristics. Since the errors are on the log scale, an RMSE of 0.425 means that a predicted revenue is typically off by a factor of about $\exp(0.43) \approx 1.5$. On the original scale, ridge regression with λ_{min} has the lowest RMSE (8.78 million USD). These errors in USD mainly come from the largest programs: the 7 FBS programs in the test set cause 71% of the squared error. The predicted revenues of the 10 new institutions (Appendix, Table 4) range from 1.7 million USD for the NAIA Division I program to 115.2 (PCR) and 142.2 million USD (ridge) for the FBS program.

5. Conclusions and Discussion

The first research question was to what extent the revenue of an athletic program can be predicted from a few underlying characteristics, compared with using all characteristics separately. Revenue can be predicted quite well. A model with three components and the athletic division explains 92% of the variance of log revenue in the test data, which is about the same as ridge regression with all 28 characteristics (91%). So a few general features of a program contain almost all the useful information. However, a prediction is still typically off by a factor of about 1.5, and the errors in USD are largest for the biggest programs. The models are therefore useful as a rough benchmark for athletic directors, but not for predicting the exact revenue of one program. The second research question was which characteristics are most closely related to revenue. Revenue is mainly related to the size of a program, and to a lesser extent to how well its coaches are paid. The division also matters a lot: FBS programs earn much more than similar programs in lower divisions.

This research has some limitations. The results only show relations, not causes. Since revenue has to cover the expenses of a program, variables such as coach salaries partly reflect the budget itself. The data are reported by the institutions themselves and describe only one point in time, and some missing salaries had to be filled in with the median. In addition, the components are formed without looking at revenue, so football, which is important for the largest programs, is not well captured by them. Finally, the models were compared on only one train/test split, so the small difference between them may change with another split.

Future research could use data from several years, repeat the split several times, and add variables that are directly related to income, such as attendance or media contracts.

References

US Department of Education (USDE). (2026). *Equity in Athletics Data Analysis*. Retrieved October 2026, from https://ope.ed.gov/athletics/#/

US Department of Education (USDE). (2025). *EADA survey: Total revenues screen instructions*. https://surveys.ope.ed.gov/athletics2025/wwwroot/documents/inst_1.pdf

Appendix

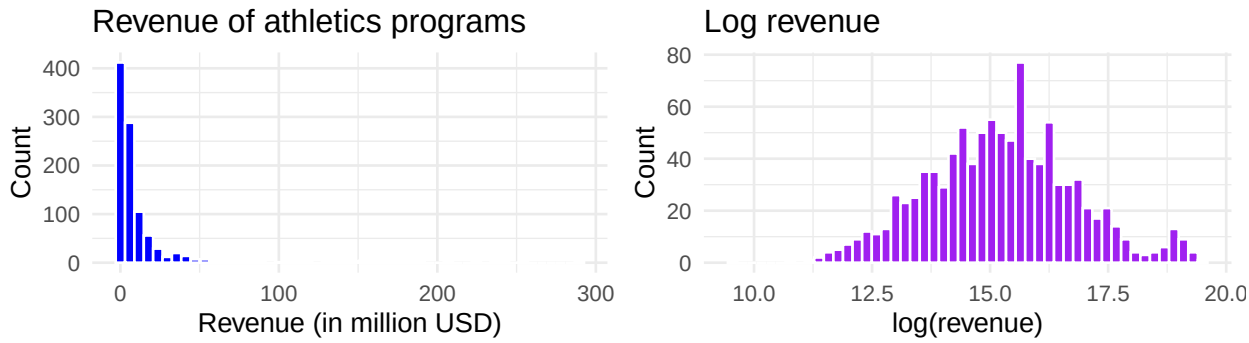


Figure 1: Distribution of revenue in million USD (left) and log revenue (right). Revenue is strongly right-skewed, log revenue is roughly symmetric.

Table 1: Eigenvalues, VAF (in %), upper bound of the permutation test and bootstrap confidence intervals of the eigenvalues for the first six components.

Component	Eigenvalue	VAF	Cumulative VAF	Permutation 97.5%	Bootstrap 95% CI
1	13.80	49.29	49.29	1.42	[13.28, 14.33]
2	2.74	9.77	59.07	1.35	[2.50, 3.09]
3	1.50	5.36	64.43	1.31	[1.39, 1.76]
4	1.23	4.40	68.83	1.27	[1.14, 1.49]
5	1.12	3.98	72.82	1.24	[1.04, 1.23]
6	1.02	3.65	76.47	1.21	[0.94, 1.10]

Table 2: Component loadings (correlations between variables and components C1 to C3) and fit per variable.

Variable	C1	C2	C3	Fit	Variable	C1	C2	C3	Fit
ugrad men	0.58	0.53	0.33	0.72	partic track women	0.78	0.15	0.24	0.70
ugrad women	0.53	0.57	0.36	0.73	partic volleyball women	0.63	-0.18	-0.20	0.47
athletes men	0.89	-0.15	-0.07	0.81	partic other men	0.75	-0.31	0.21	0.70
athletes women	0.94	-0.01	-0.02	0.88	partic other women	0.82	-0.08	0.27	0.75
teams men	0.86	-0.39	0.07	0.89	hc men	0.86	-0.38	0.05	0.89
teams women	0.92	-0.20	0.07	0.89	hc women	0.92	-0.20	0.05	0.90
partic football	0.59	0.09	-0.04	0.36	ac men	0.89	-0.03	0.07	0.79
partic bball men	0.48	-0.29	-0.48	0.55	ac women	0.87	0.03	0.14	0.79
partic bball women	0.60	-0.14	-0.39	0.53	hc salary men	0.64	0.52	-0.14	0.70
partic baseball	0.45	-0.18	-0.37	0.37	hc salary women	0.67	0.41	-0.24	0.68
partic softball	0.54	-0.10	-0.30	0.39	ac salary men	0.63	0.49	-0.26	0.70
partic soccer men	0.40	-0.51	0.14	0.44	ac salary women	0.63	0.46	-0.29	0.69
partic soccer women	0.70	-0.14	0.19	0.54	hc fulltime share	0.50	0.44	-0.21	0.49
partic track men	0.74	0.00	0.22	0.59	hc female share	0.13	0.25	0.11	0.09

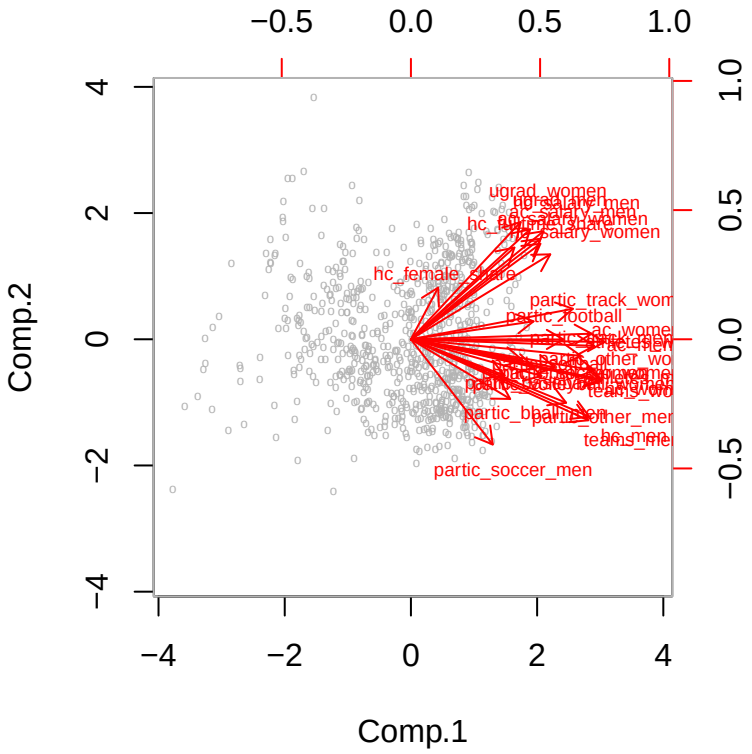


Figure 2: Biplot of components 1 and 2. Points are institutions, arrows are variables.

Table 3: Predictive performance on the test set.

Model	RMSE (log)	R^2 (log)	RMSE (million USD)
PCR	0.425	0.916	9.059
Ridge ($\lambda_{\min}$)	0.447	0.907	8.781
Ridge (λ_{1SE})	0.456	0.903	9.378

Table 4: Predicted revenue (in million USD) of the 10 institutions.

ID	Classification	PCR	Ridge
1	NCAA Division I-FBS	115.2	142.2
2	NCAA Division I-FCS	17.7	18.5
3	NCAA Division I without football	11.6	10.1
4	NCAA Division II with football	10.5	9.3
5	NCAA Division II without football	4.7	3.6
6	NCAA Division III with football	3.5	3.2
7	NCAA Division III without football	3.7	4.1
8	NAIA Division I	1.7	1.7
9	NJCAA Division I	2.2	2.7
10	CCCAA	4.4	4.7

Code

```
library(tidyverse); library(glmnet); library(boot); library(gridExtra); library(caret)

permtestPCA <- function(X, nTests = 100, alpha = 0.05, ...){ # permutation test function from lecture 5
  n <- nrow(X); m <- ncol(X); X <- scale(X)
  eigs.X <- prcomp(X)$sdev^2; eigs.Xperm <- matrix(0, m, nTests); Xperm <- X
  for (i in 1:nTests){
    for (j in 2:m) Xperm[, j] <- X[sort(runif(n), index.return = TRUE)$ix, j] # random permutation of the rows per column
    eigs.Xperm[, i] <- prcomp(Xperm)$sdev^2}
  perc.alpha <- t(apply(eigs.Xperm, 1, quantile, c(alpha/2, 1 - alpha/2)))
  plot(1:m, eigs.X, type = "b", col = "red", main = "Permutation test PCA", xlab = "Component", ylab = "Eigenvalue", ...)
  matlines(1:m, perc.alpha, type = "b", col = "blue", lty = 1)
  perc.alpha}
f2 <- function(x, d = 2) formatC(x, format = "f", digits = d)    # rounds numbers

athletics  <- read.csv("data_783996yw.csv"); prediction <- read.csv("predictions-2.csv")
dim(athletics); summary(athletics$revenue)                      # size and distribution of revenue
colSums(is.na(athletics))[colSums(is.na(athletics)) > 0]        # missing values only in the coach variables
table(no_men_teams = athletics$teams_men == 0, hc_missing = is.na(athletics$hc_men)) # no men's team -> no men's hc
min(athletics$ac_men, na.rm = TRUE)                             # ac never 0
sum(!is.na(athletics$ac_men) & is.na(athletics$ac_salary_men))  # acs but no salary (volunteers?)
table(athletics$classification); table(athletics$sector)        # some levels are very small
revenue <- ggplot(athletics, aes(revenue / 1e6)) + geom_histogram(bins = 50, fill = "blue", color = "white") +
  labs(title = "Revenue of athletics programs", x = "Revenue (in million USD)", y = "Count") + theme_minimal()
logrevenue <- ggplot(athletics, aes(log(revenue))) + geom_histogram(bins = 50, fill = "purple", color = "white") +
  labs(title = "Log revenue", x = "log(revenue)", y = "Count") + theme_minimal()
grid.arrange(revenue, logrevenue, ncol = 2)                     # very skewed, the log is more symmetric

count_vars  <- c("ugrad_men", "ugrad_women", "athletes_men", "athletes_women", "teams_men", "teams_women",
                 grep("^partic_", names(athletics), value = TRUE), "hc_men", "hc_women", "ac_men", "ac_women") # logged
salary_vars <- c("hc_salary_men", "hc_salary_women", "ac_salary_men", "ac_salary_women")                  # logged
share_vars  <- c("hc_fulltime_share", "hc_female_share")                                                 # 0 to 1, not logged
num_vars    <- c(count_vars, salary_vars, share_vars)                                                    # 28 numerical predictors
fixed_coaches <- function(x) {                                  # runs once for men's and once for women's variables
  for (s in c("men", "women")) {
    hc <- paste0("hc_", s); ac <- paste0("ac_", s); hs <- paste0("hc_salary_", s); as <- paste0("ac_salary_", s)
    no_hc <- is.na(x[[hc]]); no_ac <- is.na(x[[ac]])             # institutions without hcs / acs
    x[[hc]][no_hc] <- 0; x[[ac]][no_ac] <- 0                     # no coaches -> 0
    x[[hs]][no_hc & is.na(x[[hs]])] <- 0; x[[as]][no_ac & is.na(x[[as]])] <- 0 # coaches without salary stay na
  }
  x}
logged_vars <- function(x) { for (v in c(count_vars, salary_vars)) x[[v]] <- log1p(x[[v]]); x } # log(x + 1) as x can be 0
athletics_new  <- fixed_coaches(athletics); prediction_new <- fixed_coaches(prediction)
athletics_new$log_revenue <- log(athletics_new$revenue)         # dependent variable
athletics_new$classification <- factor(athletics_new$classification); athletics_new$sector <- factor(athletics_new$sector)
prediction_new$classification <- factor(prediction_new$classification, levels = levels(athletics_new$classification)) # same levels -> same dummies
prediction_new$sector <- factor(prediction_new$sector, levels = levels(athletics_new$sector))
athletics_new <- logged_vars(athletics_new); prediction_new <- logged_vars(prediction_new)

set.seed(123)
trainRowNumbers <- createDataPartition(athletics_new$log_revenue, p = 0.8, list = FALSE)
train_data <- athletics_new[trainRowNumbers, ]; test_data <- athletics_new[-trainRowNumbers, ]
train_medians <- sapply(train_data[num_vars], median, na.rm = TRUE) # training data only
impute <- function(x) { for (v in num_vars) x[[v]][is.na(x[[v]])] <- train_medians[v]; x }
train_data <- impute(train_data); test_data <- impute(test_data); prediction_new <- impute(prediction_new)
round(sort(sapply(train_data[num_vars], cor, y = train_data$log_revenue), decreasing = TRUE), 2) # correlations with log revenue

X_pca <- train_data[, num_vars]; X_pca_test <- test_data[, num_vars]; p <- ncol(X_pca)
fitpca <- princomp(X_pca, cor = TRUE, scores = TRUE)            # correlation matrix
eigenvalues <- fitpca$sdev^2; vaf <- eigenvalues / p; cum_vaf <- cumsum(vaf)
screeplot(fitpca, npcs = p, type = "lines", main = "Scree plot"); abline(h = 1, col = "red", lty = 2) # elbow and kaiser line
k_kaiser <- sum(eigenvalues > 1)                                # kaiser: eigenvalue > 1
k_vaf80  <- which(cum_vaf >= 0.80)[1]                           # 80% cumulative vaf
set.seed(123); perm_range <- permtestPCA(X_pca, nTests = 1000)  # permutation test
k_perm <- which(eigenvalues <= perm_range[, 2])[1] - 1          # keep pcs above the 97.5% interval bound
my_boot_pca <- function(x, ind) princomp(x[ind, ], cor = TRUE)$sdev^2 # pca on resampled rows -> eigenvalues
set.seed(123); eigs.boot <- boot(data = X_pca, statistic = my_boot_pca, R = 1000)$t
kaiser_boot <- data.frame(PC = 1:p, Eigenvalue = eigenvalues, Lower = apply(eigs.boot, 2, quantile, 0.025),
                          Upper = apply(eigs.boot, 2, quantile, 0.975))
k_kaiser_boot <- which(kaiser_boot$Lower <= 1)[1] - 1           # bootstrapped kaiser, whole confidence interval above 1
data.frame(Method = c("Kaiser", "80% VAF", "Permutation test", "Bootstrapped Kaiser"),
           PCs = c(k_kaiser, k_vaf80, k_perm, k_kaiser_boot))
n_pc <- k_perm                                                  # chosen: permutation test
vaf.boot   <- rowSums(eigs.boot[, 1:n_pc, drop = FALSE]) / p    # bootstrapped vaf of the chosen pcs
perc.alpha <- quantile(vaf.boot, c(0.025, 0.975)); round(100 * c(cum_vaf[n_pc], perc.alpha), 1)
corpca  <- cor(X_pca, fitpca$scores)[, 1:n_pc]                  # loadings
fit_var <- cbind(corpca, Fit = rowSums(corpca^2)); round(fit_var, 2) # fit per variable
biplot(fitpca, pc.biplot = TRUE, scale = 1, choices = 1:2, col = c("grey", "red"), asp = 1,
       cex = c(0.4, 0.6), xlabs = rep("o", nrow(X_pca)))      # points = institutions, arrows = variables

train_scores <- data.frame(fitpca$scores[, 1:n_pc], classification = train_data$classification,
                           sector = train_data$sector, log_revenue = train_data$log_revenue)
test_scores  <- data.frame(predict(fitpca, newdata = X_pca_test)[, 1:n_pc], # test projected on the training pca
                           classification = test_data$classification, sector = test_data$sector)
fit_pcr <- lm(log_revenue ~ ., data = train_scores); summary(fit_pcr)

form_x <- as.formula(paste("~", paste(c(num_vars, "classification", "sector"), collapse = " + ")))
y <- train_data$log_revenue
X <- model.matrix(form_x, train_data)[, -1]                     # dummies, intercept removed
X_test <- model.matrix(form_x, test_data)[, -1]; X_new <- model.matrix(form_x, prediction_new)[, -1]
x_mean <- colMeans(X[, num_vars]); x_sd <- apply(X[, num_vars], 2, sd) # z-scores with training means and sds
X[, num_vars] <- scale(X[, num_vars], x_mean, x_sd); X_test[, num_vars] <- scale(X_test[, num_vars], x_mean, x_sd)
X_new[, num_vars] <- scale(X_new[, num_vars], x_mean, x_sd)
grid <- 10^seq(-5, 2, length.out = 100)                         # grid of lambdas
plot(glmnet(X, y, alpha = 0, lambda = grid, standardize = FALSE), xvar = "lambda") # coefficient paths
set.seed(123); result.ridge.cv <- cv.glmnet(X, y, alpha = 0, lambda = grid, nfolds = 10, standardize = FALSE)
c(result.ridge.cv$lambda.min, result.ridge.cv$lambda.1se)       # lambda.min and 1se rule
result.ridge.min <- glmnet(X, y, alpha = 0, lambda = result.ridge.cv$lambda.min, standardize = FALSE)
result.ridge.1se <- glmnet(X, y, alpha = 0, lambda = result.ridge.cv$lambda.1se, standardize = FALSE)
round(cbind(result.ridge.min$beta, result.ridge.1se$beta), 3)   # coefficients per sd for numerical predictors
```

```
rmse <- function(y, yhat) sqrt(mean((y - yhat)^2))
y_test <- test_data$log_revenue; rev_test <- test_data$revenue
pred_log <- list("PCR" = predict(fit_pcr, newdata = test_scores),
                 "Ridge lambda.min" = as.numeric(predict(result.ridge.min, newx = X_test)),
                 "Ridge lambda.1se" = as.numeric(predict(result.ridge.1se, newx = X_test)))
eval_tab <- do.call(rbind, lapply(names(pred_log), function(m) { pl <- pred_log[[m]]
  data.frame(Model = m, RMSE_log = rmse(y_test, pl), R2_log = 1 - sum((y_test - pl)^2) / sum((y_test - mean(y_test))^2),
             RMSE_rev_M = rmse(rev_test, exp(pl)) / 1e6) }))   # rmse in million usd after exp()
eval_tab
err_by_class <- data.frame(classification = test_data$classification,   # where do the usd errors come from?
                           sq_err_rev = (rev_test - exp(pred_log[["Ridge lambda.min"]]))^2) %>%
  group_by(classification) %>% summarise(n = n(), share_of_squared_err = sum(sq_err_rev)) %>%
  mutate(share_of_squared_err = share_of_squared_err / sum(share_of_squared_err)) %>% arrange(desc(share_of_squared_err))
err_by_class                                                 # fbs schools cause most of the usd error

new_scores <- data.frame(predict(fitpca, newdata = prediction_new[, num_vars])[, 1:n_pc],
                         classification = prediction_new$classification, sector = prediction_new$sector)
pred_out <- data.frame(id = prediction_new$id, classification = prediction_new$classification,
                       rev_PCR_M = exp(predict(fit_pcr, newdata = new_scores)) / 1e6,          # back to usd with exp()
                       rev_Ridge_M = exp(as.numeric(predict(result.ridge.min, newx = X_new))) / 1e6)
pred_out
```